Introduction to the Theory of Computation 2026 — Midterm 1

Solutions

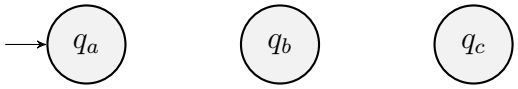
Problem 1 (15 pts). We have

$$\Sigma = \{0, 1\}.$$

(a) (5 pts) Let us consider the language

$$L_B = \{\boldsymbol{w} \mid \text{the binary value of } \boldsymbol{w} \text{ is a multiple of } 3\}.$$

Please give a DFA to recognize the language L_B with the following drafts:



That is, you should add links to connect nodes and may put some nodes as final states. Briefly explain the meaning of each state and how the transitions work. No credit will be given for answers **without an explanation**.

(b) (5 pts) On the diagram you draw in (a), please simulate the strings

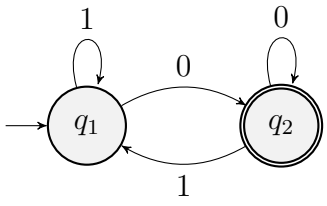
11101 and **10010** .

Note that you need to specify the strings are accepted or rejected.

(c) (5 pts) Now, we have another language

$$L_A = \{\boldsymbol{w} \mid \boldsymbol{w} \text{ ends with } 0\}$$

that can be recognized by the DFA



If we want a DFA that can recognize the language $L_A \cap L_B$, please utilize the following method

if

$$M_A = (Q_A, \Sigma, \delta_A, q_A, F_A)$$

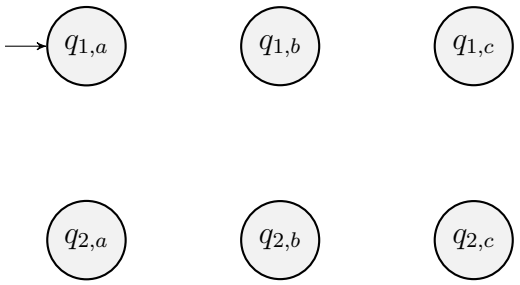
and

$$M_B = (Q_B, \Sigma, \delta_B, q_B, F_B)$$

recognize L_A and L_B respectively, the formal definition of M for $L_A \cap L_B$ is

$$\begin{aligned} Q &= \{(r_A, r_B) \mid r_A \in Q_A, r_B \in Q_B\} \\ \Sigma &\text{ is the same one} \\ \delta((r_A, r_B), a) &= (\delta_A(r_A, a), \delta_B(r_B, a)) \\ q_0 &= (q_A, q_B) \\ F &= \{(r_A, r_B) \mid r_A \in F_A \textbf{ and } r_B \in F_B\}. \end{aligned}$$

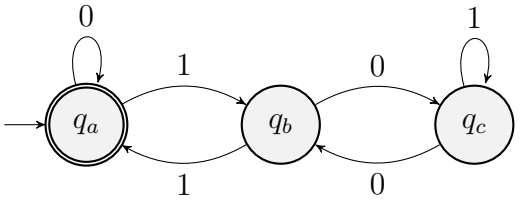
to generate the DFA with the following draft



Note that you are not required to simplify the diagram. Please answer this problem with those six specific nodes.

Solution.

(a) Please check the following diagram.



The states q_a, q_b, and q_c represent the remainders $0, 1$, and 2, respectively. The DFA processes the binary string from left to right and keeps track of the remainder modulo 3 of the value represented by the string read so far. When a new bit $b \in \{0, 1\}$ is read, the remainder r will be updated to $2r + b \mod 3$.

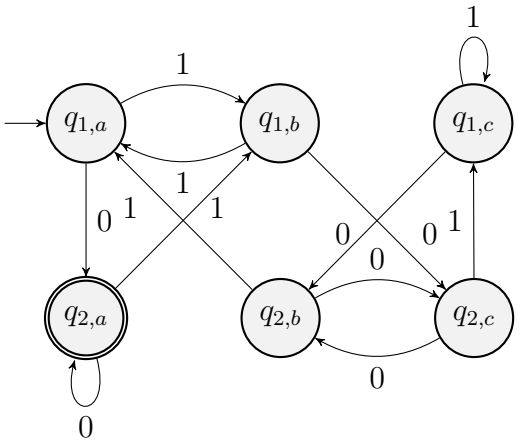
(b) For the string **11101**:

$$\to q_a \xrightarrow{1} q_b \xrightarrow{1} q_a \xrightarrow{1} q_b \xrightarrow{0} q_c \xrightarrow{1} q_c \to \text{rejected}$$

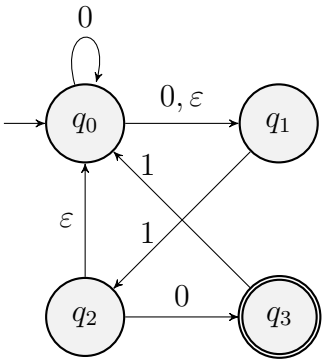
For the string **10010**:

$$\to q_a \xrightarrow{1} q_b \xrightarrow{0} q_c \xrightarrow{0} q_b \xrightarrow{1} q_a \xrightarrow{0} q_a \to \text{accepted}$$

(c) Please check the following diagram.



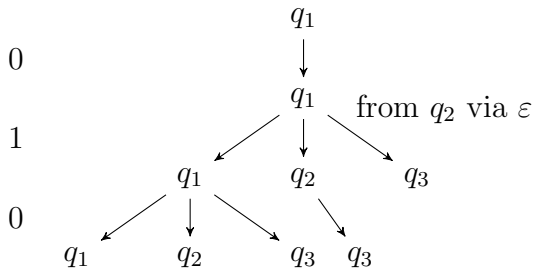
Problem 2 (60 pts). Let $\Sigma = \{0, 1\}, Q = \{q_0, q_1, q_2, q_3\}$. Consider the following NFA A.



(a) (5 pts) Please draw the computation of the NFA A on the input strings

010 and **101**.

Note that you need to conclude whether they are **accepted** or **rejected**. A computation is like the following figure (copied from Figure 1.29 in the textbook, as an illustration and not related to the NFA in this subproblem.)



Please follow the rules in the textbook. Note that we handle the ε edge **immediately**, and you have to list all possible states that can be reached once processing each input character.

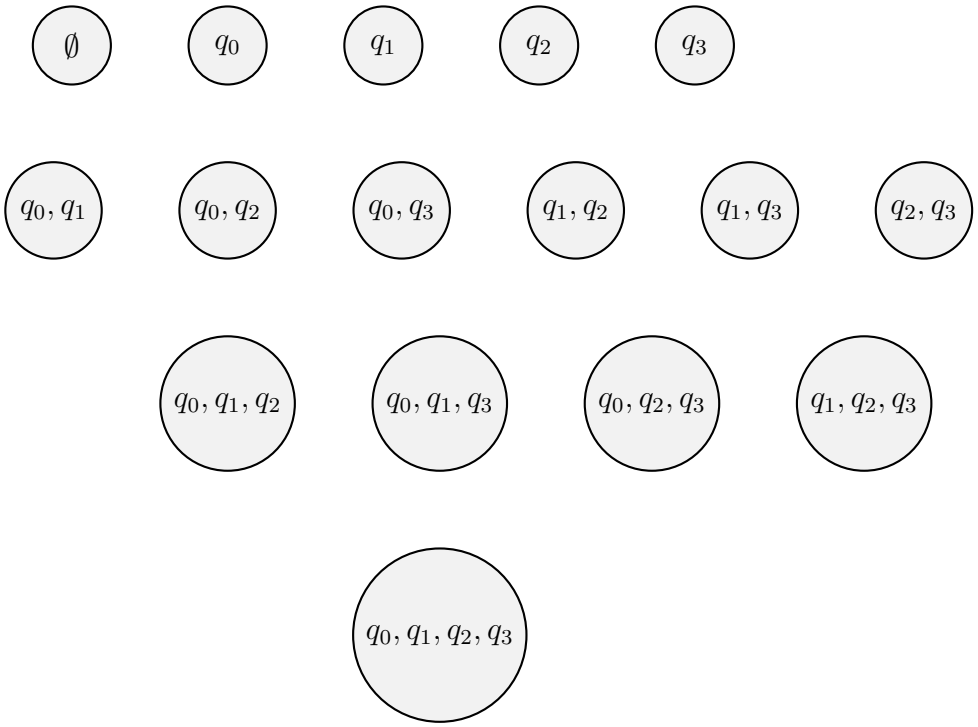
(b) (5 pts) Convert the NFA A to a DFA by the given procedure.

i. Each state is an element of the power set of Q (i.e., $P(Q)$).

ii. Started state: The state we can reach in the beginning.

iii. Accepted state: Any state including q_3 is an accepted state.

We have provided the following diagram illustrating all possible state combinations. Use this diagram for your conversion and show the result. Then remove useless states and show the resulting diagram.



(c) (10 pts) Convert the resulting DFA of the previous subproblem to a regular expression by constructing an equivalent GNFA using the following steps:

(i) Add a new started state q_s and an accepted state q_a to create the initial state diagram S_{initial}.

(ii) Remove the states in the following order (other states should already been removed in the previous subproblem):

$$\{q_0, q_1\}, \{q_0, q_1, q_2\}, \{q_0, q_1, q_3\}$$

during the conversion process to get the final state diagram S_{final}.

Please show all the details of the conversion process from S_{initial} to S_{final}. Note that you do not need to simplify the result regular expression for this subproblem.

(d) (10 pts) Convert the NFA A to a regular expression by constructing an equivalent GNFA using the following steps:

(i) Add a new started state q_s and an accepted state q_a to create the initial state diagram S_{initial}.

(ii) Remove the states $q_0, q_1, q_2, \ldots$ in **numerical order** during the conversion process to get the final state diagram S_{final}.

Please show all the details of the conversion process from S_{initial} to S_{final}. Note that you do not need to simplify the result regular expression for this subproblem.

(e) (5 pts) We would like to prove that the regular expressions obtained from the DFA and the NFA are equivalent. This is a complicated task, so we split it into several subproblems. To begin, we prove the following properties:
For any regular expression R, we have

(a) $R^*R^* = R^*$,

(b) $(R^*)^* = R^*$.

First, we prove (a). For any $x \in R^*$, since $\varepsilon \in R^*$, we have $x = x\varepsilon \in R^*R^*$, which implies that $R^* \subseteq R^*R^*$. On the other hand, let $x \in R^*R^*$. Then $x = x_1x_2$ for some $x_1, x_2 \in R^*$. By the definition of R^*, we can write

$$x_1 = x_{11} \cdots x_{1k_1} \text{ with } k_1 \geq 0 \text{ and } x_{1i} \in R, \text{ for all } i,$$
$$x_2 = x_{21} \cdots x_{2k_2} \text{ with } k_2 \geq 0 \text{ and } x_{2i} \in R, \text{ for all } i.$$

It follows that

$$x_1x_2 = x_{11} \cdots x_{1k_1}x_{21} \cdots x_{2k_2} \in R^*.$$

Thus, $R^*R^* \subseteq R^*$. Combining both inclusions, we conclude that $R^*R^* = R^*$. Now, please use a similar scheme (with the same details) to show that $(R^*)^* = R^*$.

(f) (10 pts) Since $0^*(0 \cup \varepsilon)$ and 0^* are equivalent, the regular expression obtained from the NFA can be simplified to the following regular expression:

$$R_1 = (0^*1)^+0(1(0^*1)^+0)^*.$$

Note that for convenience, we let R^+ be shorthand for RR^*. (copied from Page 65 in the textbook.) In fact, R_1 can be further simplified to:

$$R_2 = (0^*1)^+0.$$

Now, we would like to show that R_1 and R_2 are equivalent. We still prove that $R_1 \subseteq R_2$ and $R_2 \subseteq R_1$. Please show that $R_2 \subseteq R_1$ first as it is easier. Next, for $R_1 \subseteq R_2$, complete the following derivation by filling in the missing intermediate steps.
We have

$$\begin{aligned} R_1 &= (0^*1)^+0(\varepsilon \cup (1(0^*1)^+0) \cup (1(0^*1)^+0)^2 \cup \cdots) \\ &\vdots \\ &= (0^*1)^+(01(0^*1)^+)^*0. \end{aligned}$$

Since ..., we can use the results in 2(e) to have .... Thus,

$$R_1 \cdots \subseteq \cdots R_2.$$

(g) (15 pts) Now we use the results in 2(f) to show that the regular expressions obtained from the DFA and the NFA are equivalent.
First, we simplify the resulting regular expressions from using the DFA and the NFA. Since $1 \cup 00^*1 = 0^*1$, and by the equivalence proved in 2(f), the regular expressions obtained from the DFA and the NFA can be simplified as follows:

$$\text{DFA: } R_{\text{DFA}} = (0^*1^+0)^+$$
$$\text{NFA: } R_{\text{NFA}} = (0^*1)^+0$$

We still use the same strategy of proving

$$R_{\text{DFA}} \subseteq R_{\text{NFA}} \text{ and } R_{\text{NFA}} \subseteq R_{\text{DFA}}$$

to get $R_{\text{DFA}} = R_{\text{NFA}}$. We start with proving $R_{\text{DFA}} \subseteq R_{\text{NFA}}$. Because

$$R_{\text{DFA}} = \bigcup_{n=1}^{\infty} (0^*1^+0)^n,$$

it is sufficient to show that

$$(0^*1^+0)^n \subseteq (0^*1)^+0 \quad \text{for all } n \in \mathbb{N}. \tag{i}$$

We prove the above statement by induction. When $n = 1$,

$$0^*1^+0 = 0^*11^*0 = 0^*1(\varepsilon 1)^*0 \subseteq 0^*1(0^*1)^*0 = (0^*1)^+0.$$

Assume that $(0^*11^*0)^n \subseteq (0^*1)^+0$ holds when $n = k$. Then for $n = k + 1$, we have

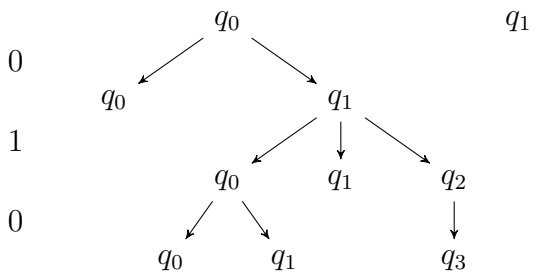
$$\begin{aligned}
(0^*1^+0)^{k+1} &= 0^*1^+0(0^*1^+0)^k \\
&\subseteq 0^*1^+0(0^*1)^+0 && \text{(ii)} \\
&= (0^*1)(\varepsilon 1)^*(00^*1)(0^*1)^*0 \\
&\subseteq (0^*1)(0^*1)^*(0^*1)(0^*1)^*0 \\
&\subseteq 0^*1(0^*1)^*0 && \text{(iii)} \\
&= (0^*1)^+0,
\end{aligned}$$

where (ii) is from the induction assumption and (iii) is from 2(e). By mathematical induction, this proves (i).
Your task is to prove the other direction (i.e., $R_{\text{NFA}} \subseteq R_{\text{DFA}}$) by the same scheme. Specifically, you must use mathematical induction on n for $(0^*1)^n0$.
Hint: For simplicity of writing the proof, in one place, you should separately show that

$$0^*10^*1^+0 \subseteq (0^*1^+0)^+.$$

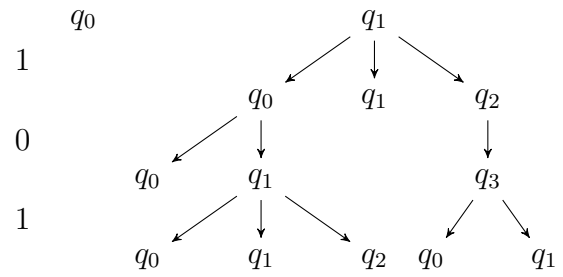
Solution.

(a) For the input string **010**:



This NFA **accepts** the input string **010**.

For the input string **101**:



This NFA **rejects** the input string **101**.

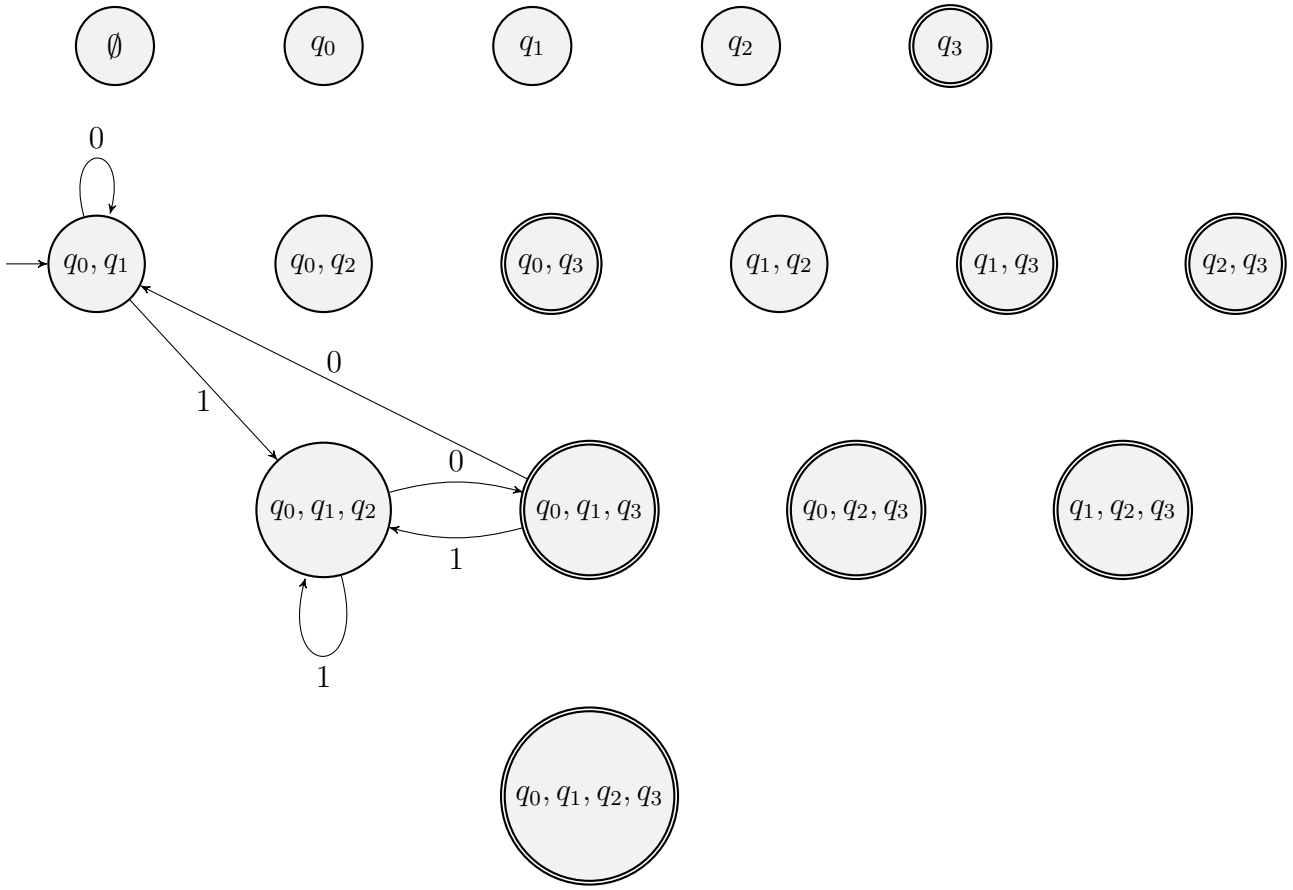
(b) Step 1: We have the combination of the states as

$$\emptyset, \{q_0\}, \{q_1\}, \{q_2\}, \{q_3\}, \{q_0, q_1\}, \{q_0, q_2\}, \{q_0, q_3\}, \{q_1, q_2\}, \{q_1, q_3\}, \{q_2, q_3\},$$
$$\{q_0, q_1, q_2\}, \{q_0, q_1, q_3\}, \{q_0, q_2, q_3\}, \{q_1, q_2, q_3\}, \{q_0, q_1, q_2, q_3\}$$

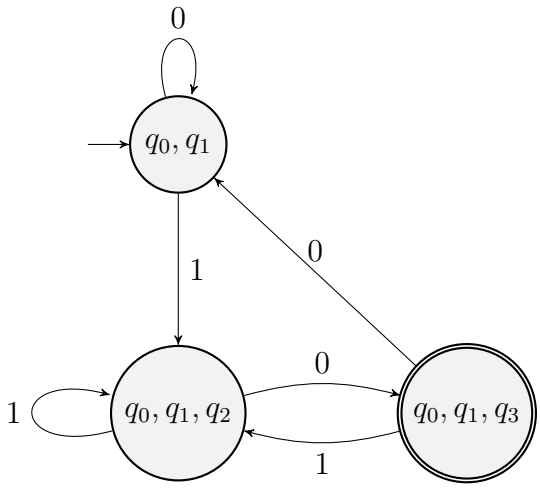
Step 2: We now follow the procedure outlined in our slides "chap1_NFA3.pdf":

i. Start state: $\{q_0, q_1\}$.

ii. Accept state: Any state that includes q_3 is considered an accept state.

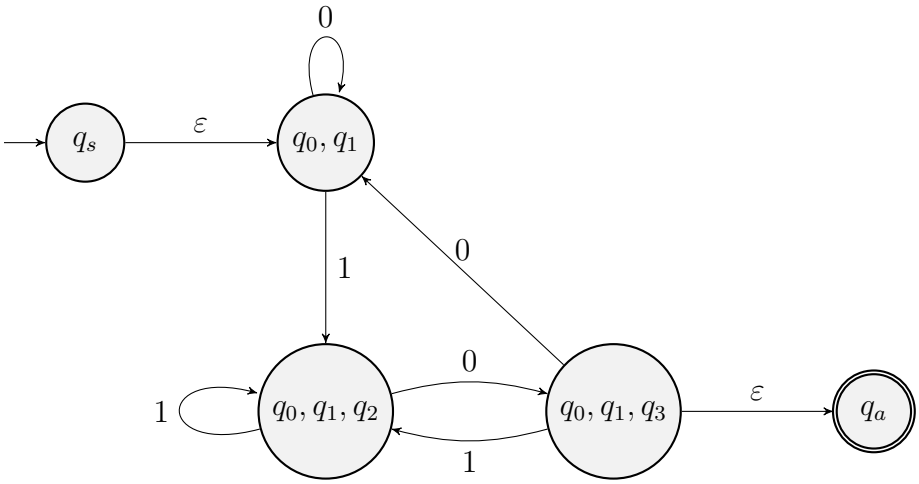


Step 3: After we remove the unused states, we have

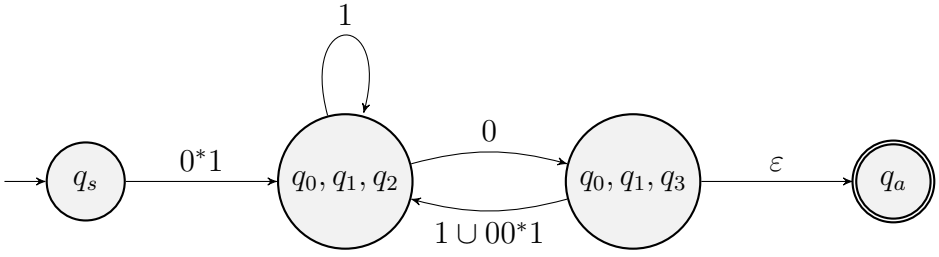


(c) We have the following steps:

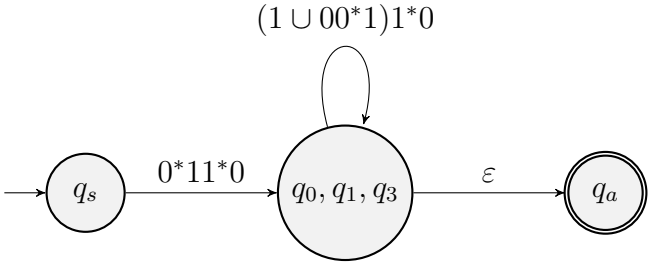
Step 1. Add new start q_s and accept q_a states.



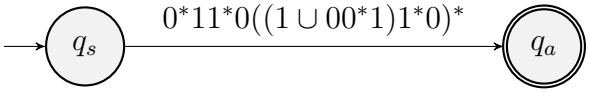
Step 2. Remove $\{q_0, q_1\}$.



Step 3. Remove $\{q_0, q_1, q_2\}$.

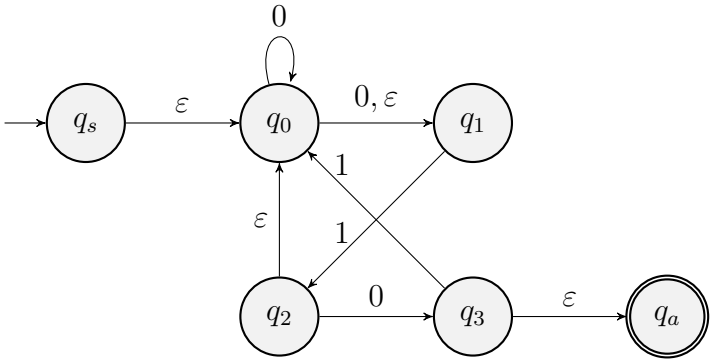


Step 4. Remove $\{q_0, q_1, q_3\}$.

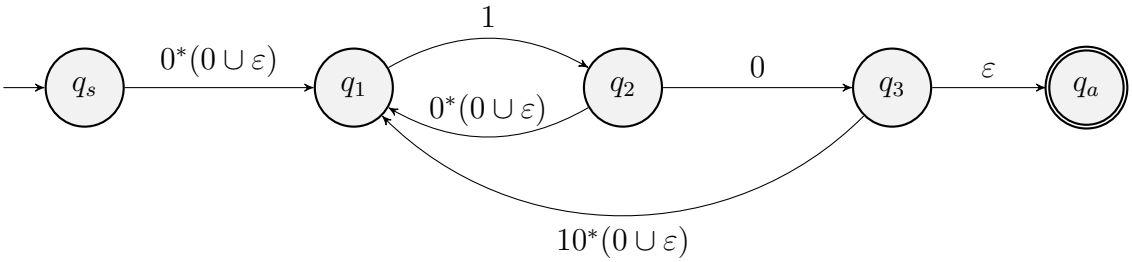


(d) We have the following steps:

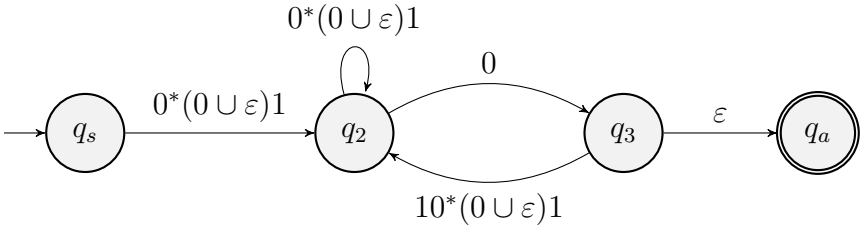
Step 1. Add new start q_s and accept q_a states.



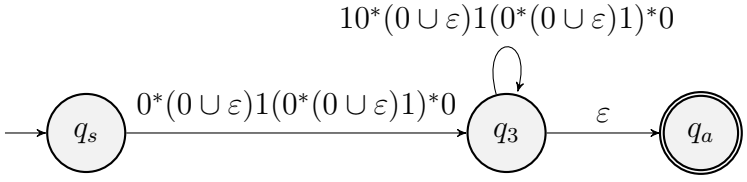
Step 2. Remove q_0.



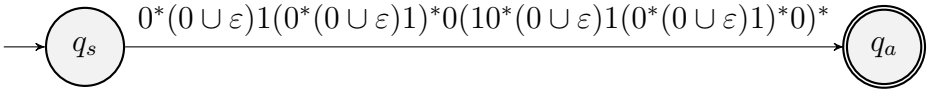
Step 3. Remove q_1.



Step 4. Remove q_2.



Step 5. Remove q_3.



(e) By the definition of $(R^*)^*$, it is clear that $R^* \subseteq (R^*)^*$. On the other hand, for any $x \in (R^*)^*$,

$$x = x_1 \cdots x_n \text{ with } n \geq 0 \text{ and } x_i \in R^*, \text{ for all } i.$$

As $x_i \in R^*$ for all i, we can write

$$\begin{aligned}
x_1 &= x_{11} \cdots x_{1k_1} \text{ with } k_1 \geq 0 \text{ and } x_{1j} \in R, \text{ for all } j, \\
x_2 &= x_{21} \cdots x_{2k_2} \text{ with } k_2 \geq 0 \text{ and } x_{2j} \in R, \text{ for all } j, \\
&\vdots \\
x_n &= x_{n1} \cdots x_{nk_n} \text{ with } k_n \geq 0 \text{ and } x_{nj} \in R, \text{ for all } j,
\end{aligned}$$

It follows that

$$x = x_1 \cdots x_n = x_{11} \cdots x_{1k_1} \cdots x_{n1} \cdots x_{nk_n} \in R^*.$$

Thus, $(R^*)^* \subseteq R^*$. Combining both inclusions, we conclude that $(R^*)^* = R^*$.

(f) Since $\varepsilon \in (1(0^*1)^+0)^*$, it follows that $R_2 \subseteq R_1$. For the other direction, we have

$$\begin{aligned}
R_1 &= (0^*1)^+0(\varepsilon \cup (1(0^*1)^+0) \cup (1(0^*1)^+0)^2 \cup \cdots) \\
&= (0^*1)^+(0 \cup 0(1(0^*1)^+0) \cup 0(1(0^*1)^+0)^2 \cup \cdots) \\
&= (0^*1)^+(0 \cup (01(0^*1)^+)0 \cup (01(0^*1)^+)^20 \cup \cdots) \\
&= (0^*1)^+(\varepsilon \cup (01(0^*1)^+) \cup (01(0^*1)^+)^2 \cup \cdots)0 \\
&= (0^*1)^+(01(0^*1)^+)^*0.
\end{aligned}$$

Since $01 \in (0^*1)^*$, and $(0^*1)^+ \subseteq (0^*1)^*$, we can derive

$$(01(0^*1)^+)^* \subseteq (0^*1)^* \tag{iv}$$

from the properties we proved in 2(e). Thus,

$$\begin{aligned}
R_1 &= (0^*1)^+(01(0^*1)^+)^*0 \\
&= (0^*1)(0^*1)^*(01(0^*1)^+)^*0 \\
&\subseteq (0^*1)(0^*1)^*(0^*1)^*0 && \text{(v)} \\
&= (0^*1)(0^*1)^*0 && \text{(vi)} \\
&= (0^*1)^+0,
\end{aligned}$$

where (v) is from (iv) and (vi) is from 2(e). Hence, R_1 and R_2 are equivalent.

(g) Because

$$R_{\text{NFA}} = \bigcup_{n=1}^{\infty} (0^*1)^n 0,$$

it is sufficient to show that

$$(0^*1)^n 0 \subseteq (0^*1^+0)^+ \quad \text{for all } n \in \mathbb{N}. \tag{vii}$$

We prove the above statement by induction on n. When $n = 1$,

$$0^*10 \subseteq 0^*1^+0 \subseteq (0^*1^+0)^+.$$

Assume that $(0^*1)^n0 \subseteq (0^*1^+0)^+$ holds when $n = k$. Then for $n = k + 1$, we have

$$\begin{aligned}(0^*1)^{k+1}0 &= 0^*1(0^*1)^k0 \\ &\subseteq 0^*1(0^*1^+0)^+ && \text{(viii)} \\ &= 0^*10^*1^+0(0^*1^+0)^*, && \text{(ix)}\end{aligned}$$

where (viii) is from the induction assumption. We argue that

$$0^*10^*1^+0 \subseteq (0^*1^+0)^+ \tag{x}$$

by

$$\begin{aligned}0^*10^*1^+0 &= 0^*1(\varepsilon \cup 0^+)1^+0 \\ &= 0^*11^+0 \cup 0^*10^+1^+0 \\ &= 0^*11^+0 \cup 0^*100^*1^+0 \\ &\subseteq (0^*1^+0)^+ \cup (0^*1^+0)^+ \\ &= (0^*1^+0)^+.\end{aligned}$$

With (x),

$$\begin{aligned}\text{(ix)} &\subseteq (0^*1^+0)^+(0^*1^+0)^* \\ &= (0^*1^+0)(0^*1^+0)^*(0^*1^+0)^* \\ &= (0^*1^+0)^+, && \text{(xi)}\end{aligned}$$

where (xi) is from 2(e). By mathematical induction, this proves (vii).

Problem 3 (25 pts). Let

$$\Sigma_2 = \left\{ \begin{bmatrix}0\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}, \begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}1\\1\end{bmatrix} \right\}.$$

Each symbol in Σ_2 is a column containing two bits. A string

$$w = \begin{bmatrix}a_1\\b_1\end{bmatrix} \begin{bmatrix}a_2\\b_2\end{bmatrix} \cdots \begin{bmatrix}a_n\\b_n\end{bmatrix} \in \Sigma_2^*$$

represents two binary numbers: the top row is $a_1a_2 \cdots a_n$, and the bottom row is $b_1b_2 \cdots b_n$. Both rows are read from left to right, with the leftmost bit being the most significant bit. For example,

$$\begin{bmatrix}0\\0\end{bmatrix} \begin{bmatrix}0\\1\end{bmatrix} \begin{bmatrix}1\\1\end{bmatrix} \begin{bmatrix}0\\0\end{bmatrix}$$

has top row $0010_2 = 2$ and bottom row $0110_2 = 6$. Consider the language

$$C = \{w \in \Sigma_2^* \mid \text{the bottom row of } w \text{ is three times the top row}\}.$$

Please answer the following questions.

(a) (10 pts) Consider the input string

$$\begin{bmatrix}0\\0\end{bmatrix} \begin{bmatrix}0\\1\end{bmatrix} \begin{bmatrix}1\\1\end{bmatrix}.$$

Suppose that only the first two symbols have been read. Let x denote the binary number represented by the top row of the prefix read so far, and let y denote the binary number represented by the bottom row. In this example,

$$x = (00)_2 = 0 \text{ and } y = (01)_2 = 1.$$

Define the difference between y and three times x as

$$d = y - 3x.$$

Therefore, for the prefix read so far,

$$d = 1 - 3(0) = 1.$$

Now suppose that the next reading symbol is

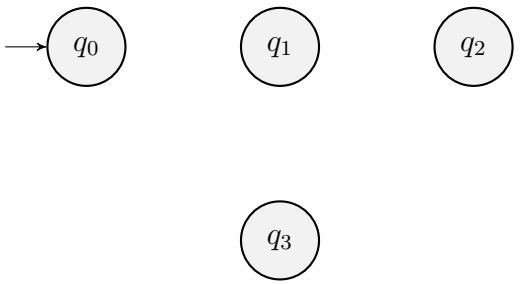
$$\begin{bmatrix} a \\ b \end{bmatrix},$$

where a is the top bit and b is the bottom bit. Appending bit a to the right of x and bit b to the right of y changes their values to x' and y', respectively. So the new difference between y' and three times x' is

$$d' = y' - 3x'.$$

You can see that we aim to check if $d = 0$ after finishing reading the input. Please express d' in terms of $d,\ a,$ and b.

(b) (15 pts) Use the change from d to d' you derive in (a) to construct a DFA that recognizes C with following draft



Clearly explain the meaning of every state and why your DFA is correct.

Solution.

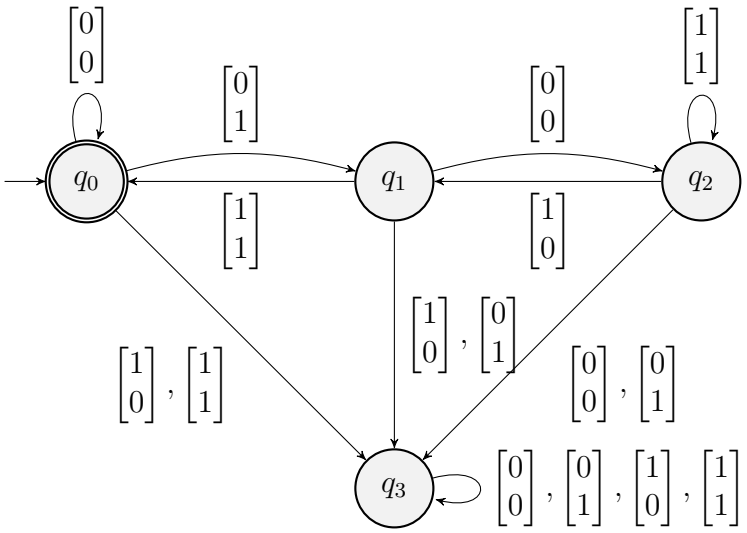
(a) Appending new bits a and b to the right of x and y, respectively, gives

$$x' = 2x + a \text{ and } y' = 2y + b.$$

Hence, their new difference is

$$\begin{aligned} d' &= y' - 3x' \\ &= (2y + b) - 3(2x + a) \\ &= 2(y - 3x) + b - 3a \\ &= 2d + b - 3a. \end{aligned}$$

(b) Please check the following diagram.



For any prefix of the input string that has been read, let x and y be the binary numbers on the top and bottom rows, respectively, and let $d = y - 3x$. The states have the following meanings:

- q_0 represents $d = 0$;
- q_1 represents $d = 1$;
- q_2 represents $d = 2$; and
- q_3 represents $d < 0$ or $d \geq 3$.

The state q_3 is a fail state. If $d \leq -1$, and for every input symbol we have

$$d' = 2d + b - 3a \leq 2(-1) + 1 = -1 < 0.$$

Similarly, if $d \geq 3$, then

$$d' = 2d + b - 3a \geq 2(3) - 3 = 3.$$

Thus, once the DFA reaches q_3, it can never return to q_0, q_1 or q_2.